COLORADO

Chief Information Officer & Executive Director

Governor's Office of Information Technology

TS-DBS-003: Database Patch Management

Document ID	TS-DBS-003
Title	Database Patch Management
Executive Sponsor	Database Services Management, DBS Manager
Issuing Authority	Governor's Office of Information Technology (OIT)
Applies to	OIT and OIT-supported agencies, including employees of OIT, agencies supported by OIT, and vendors contracted by OIT or its stakeholders.
Instrument Type	Standard
Version	5.0
Effective Date	01/27/2021
Last Updated	07/01/2026

Purpose

This standard is issued under the authority of the Governor's Office of Information Technology (OIT) (C.R.S. 24-37.5, et seq.) and the Office of Enterprise Architecture (Rule 8 CCR 1501-7). It communicates the standards OIT has developed for providing DBMS platform patching requirements.

Standards are based on a best-of-value decision assessment involving multiple factors, such as stakeholder needs, hardware configurations serviced and supported by OIT, and operational and security best practices.

Scope

These standards apply to OIT and OIT-supported agencies. This standard shall be used by employees of OIT, agencies supported by OIT, and vendors contracted by OIT or its stakeholders as a means to enforce technology standardization and assist in making the usage, support, and/or purchase of technologies more consistent and efficient.

Standards instituted for use by approval of the OIT Architecture Review Board (ARB) govern purchase and deployment decisions deemed "go-forward" or "net-new", and are not intended as a retroactively enforceable mandate for immediate conformance, especially where cost implications may be prohibitive. This standard shall in no way require OIT to support any products and services not approved by OIT.

For the purpose of enforcement, this standard is rated CAT 1 according to the Enforcement Category Chart.

Roles and Responsibilities

Enterprise Architecture - Chief Enterprise Architect

- Chairs and conducts business of the Architecture Review Board (ARB).
- Updates effective date and/or last review date.
- Publishes the ARB-approved Standard to the OIT public-facing website.

Architecture Review Board (ARB)

- Engages stakeholders and subject matter experts to determine the standard.
- Reviews/approves the Standard (this document).

Database Services Management - DBS Manager

- Owner of Standard.
- Engages stakeholders and subject matter experts to determine the standard.
- Updates document standards and version number.
- Presents to the Architecture Review Board for approval and adoption.

All OIT staff are responsible for being aware of the existence of this (and all) standard(s) and its purposes of authority and governance in support of OIT's statutorily mandated mission in service of Colorado State Government affairs.

Standards

The Database Administrator must contact the Infrastructure Operations team, including the Infrastructure Operations Manager, and advise of the proposed patch deployment dates for development, test, and production to confirm these dates.

Patching levels are to be maintained at the highest level available for the database technology employed, particularly as it pertains to security updates.

Patching Requirements

Prior to patching production, patches will be applied to non-production environments, tested and confirmed by the Infrastructure Operations team and Technical Owner, in the following order:

- Sandbox*
- Development*
- Test/Staging/UAT/Training*
- Production

*In the absence of an existing environment or disparity between non-production environments and production, the next server in sequence may be substituted.

The Database Administrator must contact the OIT Service Desk (SD) with information on the upcoming maintenance using the Service Desk Outage Notification template, to notify the business, and must send an email reminder notification to the Infrastructure Operations team to confirm nothing has changed that would prevent the planned database unavailability.

Post-Patching Requirements

Upon completion of database patching, if any errors or warnings are discovered, the cause must be investigated and, if necessary, the patching process rolled back and repeated with a resolution in place. Completed patches must not contain any errors or warnings.

The Database Administrator must initiate communication with the Infrastructure Operations team and Database Administrator group to communicate patching success or failure. If a failure occurs, the Database Administrator will work with the Infrastructure Operations team to either reschedule the patch or provide a completion time if it can be resolved the same day.

Revision History

This standard is to be reviewed every 12 months by the document owner and remains in effect until otherwise noted.

Variance and Exception Process

Requests for variances from any provision of this standard must be requested using the Technical Standard End User Exception Request Procedure.

The Exception Request process addresses business or technical reasons for the inability to adhere to one or more requirements in this standard. Requests for exceptions should include compensating controls used to mitigate any deviation from the standard requirement. The Secure Configuration Exception Request form must be submitted to oit_ciso@state.co.us. Exception requests will be processed and recommended for approval, conditional approval, or denial.

References

- State documents and policies
- C.R.S. 24-37.5-101, et seq.
- OIT documents and policies
- Enforcement Category Chart
- OIT Database Backup and Recovery Technical Policy
- Database Services documents and policies
- IAW Technology Standards

Definitions

- Standard: A rule that specifies a particular course of action or response to a given situation. Standards are mandatory directives to carry out management's policies and are used to measure compliance with policies.
- Production Database: State Agency database used for production purposes to conduct State business.
- Non-Production Database: State Agency database used for non-production purposes such as development, testing, staging, demonstration, or training.
- Business: Employees of supported State agencies, and their respective stakeholders, who use DBS-supported databases.
- Infrastructure Operations Team: Enterprise Application development team responsible for application development.

- OEA (Office of Enterprise Architecture): Formed as an office under OIT by Governor's Rule 8 CCR 1501-7.
- OIT (Office of Information Technology): Created and governed by C.R.S. 24-37.5, et seq.
- Exception request: A document requesting an exception to a recognized standard, requiring a business or technical justification for granting the exception, subject to review and approval by an authority in the area.